Machine Learning Task 6 - Spam Detection

Aim

To develop a model that can automatically classify SMS messages as spam or ham by converting text data into numerical features using TF-IDF and applying Lasso (L1 regularization) to perform feature selection and reduce unnecessary features.

Dataset

The SMS Spam Collection dataset consists of thousands of SMS messages labeled as either spam or ham. The dataset contains two main columns: the message text and its corresponding label. The labels are converted into numerical form where spam is represented as 1 and ham as 0, making it suitable for machine learning models.

Methodology

Initially, the dataset is preprocessed by selecting relevant columns and handling missing values. The text data is then transformed into numerical form using TF-IDF vectorization, where each unique word becomes a feature. This results in a high-dimensional dataset with thousands of features.

To handle this, Logistic Regression with L1 regularization (Lasso) is applied. Lasso reduces the complexity of the model by shrinking less important feature coefficients to zero, effectively removing them. The model is trained using different alpha values (0.01, 0.1, and 1) to observe how regularization strength affects the number of selected features. The number of non-zero coefficients is counted to determine how many features are retained.

Results

After applying TF-IDF, a total of **9444 features** were generated. When Lasso was applied:

- For alpha = 0.01, **405 features** were selected, meaning most features were still retained.
- For alpha = 0.1, **341 features** were selected, showing a balanced reduction.
- For alpha = 1, only **101 features** were selected, indicating strong feature elimination.

The percentage reduction in features for alpha = 0.1 was approximately **95.94%**, which shows that most irrelevant features were successfully removed. The graph plotted between alpha values and number of selected features clearly shows a decreasing trend, indicating that higher alpha values lead to fewer selected features.

Conclusion

The experiment demonstrates that TF-IDF effectively converts text data into numerical form, but it results in a large number of features. Lasso (L1 regularization) successfully reduces this high-dimensional data by eliminating irrelevant features and retaining only the most important ones. As the alpha value increases, the number of selected features decreases, making the model simpler and more efficient. This approach is highly useful in spam detection and other text classification tasks.